

1.6.2b:

Solution: Since there are 3 vectors in the set and $\dim \mathbb{R}^3 = 3$ it suffices to check for linear independence by Corollary 2:

$$A = \begin{pmatrix} 2 & 0 & 6 \\ -4 & 3 & 0 \\ 1 & -1 & -1 \end{pmatrix} \xrightarrow{2_1 \rightarrow \frac{1}{2}r_1} \begin{pmatrix} 1 & 0 & 3 \\ -4 & 3 & 0 \\ 1 & -1 & -1 \end{pmatrix} \xrightarrow[r_3 \rightarrow r_3 - r_1]{r_2 \rightarrow r_2 + 4r_1} \begin{pmatrix} 1 & 0 & 3 \\ 0 & 3 & 12 \\ 0 & -1 & -4 \end{pmatrix} \xrightarrow{r_3 \rightarrow 3r_3 + r_2} \begin{pmatrix} 1 & 0 & 3 \\ 0 & 3 & 12 \\ 0 & 0 & 0 \end{pmatrix}$$

thus $\text{rank}(A) = 2 < 3$ and hence the set is linearly dependent and thus not a base for $\mathbb{R}^3$.

1.6.3b:

Solution: Using the fact that $P_2(\mathbb{R}) \cong \mathbb{R}^3$ to write the vectors as column vectors in $\mathbb{R}^3$ and then check for linear independence this way:

$$A = \begin{pmatrix} 1 & 3 & 0 \\ 2 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix} \xrightarrow[r_3 \rightarrow r_3 - r_1]{r_2 \rightarrow r_2 - 2r_1} \begin{pmatrix} 1 & 3 & 0 \\ 0 & -6 & 1 \\ 0 & -2 & 1 \end{pmatrix} \xrightarrow{r_3 \rightarrow 3r_3 - r_2} \begin{pmatrix} 1 & 3 & 0 \\ 0 & -6 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

thus $\text{rank}(A) = 3$ and so the set is linearly independent and thus a basis by Corollary 2.

1.6.9:

Solution:

$$\begin{pmatrix} 1 & 0 & 0 & 0 & | & a_1 \\ 1 & 1 & 0 & 0 & | & a_2 \\ 1 & 1 & 1 & 0 & | & a_3 \\ 1 & 1 & 1 & 1 & | & a_4 \end{pmatrix} \xrightarrow[r_4 \rightarrow r_4 - r_1]{r_2 \rightarrow r_2 - r_1, r_3 \rightarrow r_3 - r_1} \begin{pmatrix} 1 & 0 & 0 & 0 & | & a_1 \\ 0 & 1 & 0 & 0 & | & a_2 - a_1 \\ 0 & 1 & 1 & 0 & | & a_3 - a_1 \\ 0 & 1 & 1 & 1 & | & a_4 - a_1 \end{pmatrix} \xrightarrow[r_4 \rightarrow r_4 - r_2]{r_3 \rightarrow r_3 - r_2} \begin{pmatrix} 1 & 0 & 0 & 0 & | & a_1 \\ 0 & 1 & 0 & 0 & | & a_2 - a_1 \\ 0 & 0 & 1 & 0 & | & a_3 - a_2 \\ 0 & 0 & 1 & 1 & | & a_4 - a_2 \end{pmatrix} \xrightarrow{r_4 \rightarrow r_4 - r_3} \begin{pmatrix} 1 & 0 & 0 & 0 & | & a_1 \\ 0 & 1 & 0 & 0 & | & a_2 - a_1 \\ 0 & 0 & 1 & 0 & | & a_3 - a_2 \\ 0 & 0 & 0 & 1 & | & a_4 - a_3 \end{pmatrix}$$

Thus,

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = a_1 \vec{u}_1 + (a_2 - a_1) \vec{u}_2 + (a_3 - a_2) \vec{u}_3 + (a_4 - a_3) \vec{u}_4$$

2.1.2:

Solution: T is of the form L_A for

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

To show T is linear, first consider $\vec{u} = (a_1, a_2, a_3)$ and $\vec{v} = (b_1, b_2, b_3)$. Then

$$T(\vec{u} + \vec{v}) = \begin{pmatrix} (a_1 + b_1) - (a_2 + b_2) \\ 2(a_3 + b_3) \end{pmatrix} = \begin{pmatrix} a_1 - a_2 \\ 2a_3 \end{pmatrix} + \begin{pmatrix} b_1 - b_2 \\ 2b_3 \end{pmatrix} = T(\vec{u}) + T(\vec{v})$$

$$T(c\vec{u}) = \begin{pmatrix} c(a_1 - a_2) \\ 2ca_3 \end{pmatrix} = c \begin{pmatrix} a_1 - a_2 \\ 2a_3 \end{pmatrix} = cT(\vec{u})$$

and so T is linear. To find a basis for $N(T)$ we see

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \Rightarrow \begin{cases} x_1 - x_2 = 0 \Rightarrow x_1 = x_2 \\ 2x_3 = 0 \Rightarrow x_3 = 0 \end{cases} \Rightarrow X = \begin{pmatrix} x_1 \\ x_1 \\ 0 \end{pmatrix} \Rightarrow \text{Span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\} = N(T)$$

and so $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}$ is a basis for $N(T)$. For $R(T)$ we have that

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right\}$$

is a basis for $R(T)$ since they are the vectors corresponding to the pivot columns of $\text{RREF}(A)$. Thus

$$\text{nul}(T) + \text{rank}(T) = 1 + 2 = 3 = \dim \mathbb{R}^3$$

verifying the dimension theorem. Since $N(T) \neq \{0\}$, T is not one-to-one by Thm2.4. T is onto since $\text{rank}(A) = 2 = \dim \mathbb{R}^2$.

2.1.3:

Solution: T is of the form L_A for

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 2 & -1 \end{pmatrix}.$$

To see that T is linear consider $\vec{u}, \vec{v} \in \mathbb{R}^2$ such that $\vec{u} = (a_1, a_2), \vec{v} = (b_1, b_2)$. Then,

$$T(\vec{u} + \vec{v}) = \begin{pmatrix} a_1 + b_1 + a_2 + b_2 \\ 0 \\ 2(a_1 + b_1) - (a_2 + b_2) \end{pmatrix} = \begin{pmatrix} a_1 + a_2 \\ 0 \\ 2a_1 - a_2 \end{pmatrix} + \begin{pmatrix} b_1 + b_2 \\ 0 \\ 2(b_1 - b_2) \end{pmatrix} = T(\vec{u}) + T(\vec{v})$$

$$T(c\vec{u}) = \begin{pmatrix} c(a_1 + a_2) \\ 0 \\ 2ca_1 - ca_2 \end{pmatrix} = c \begin{pmatrix} a_1 + a_2 \\ 0 \\ 2a_1 - a_2 \end{pmatrix} = cT(\vec{u})$$

and so T is linear. To find a basis for $N(T)$ we have

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 2 & -1 \end{pmatrix} \xrightarrow{r_2 \rightarrow r_2 - 2r_1} \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 0 & -3 \end{pmatrix} \Rightarrow \begin{cases} -3x_2 = 0 \Rightarrow x_2 = 0 \\ x_1 + x_2 = 0 \Rightarrow x_1 = -x_2 \end{cases} \Rightarrow X = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

and hence $N(T) = \{0\}$. For $R(T)$ we know from above that the first and second columns are pivot columns and so

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}$$

is a basis for $R(T)$. Since $N(T) = \{0\}$, T is one-to-one and because $\text{rank}(A) = 2 < 3 = \dim \mathbb{R}^3$ T is not onto.

2.1.9:

Solution:

(a) T is not linear since for $\vec{u} = (a_1, a_2)$ and $\vec{v} = (b_1, b_2)$ we have

$$T(\vec{u} + \vec{v}) = \begin{pmatrix} 1 \\ a_2 + b_2 \end{pmatrix} \neq \begin{pmatrix} 2 \\ a_2 + b_2 \end{pmatrix} = T(\vec{u}) + T(\vec{v})$$

(b) T is not linear since for $\vec{u} = (a_1, a_2)$ and $c \in \mathbb{R}$ we have

$$T(c\vec{u}) = \begin{pmatrix} ca_1 \\ (ca_2)^2 \end{pmatrix} \neq \begin{pmatrix} ca_1 \\ c(a_1)^2 \end{pmatrix} = cT(\vec{u})$$

2.1.15:

Solution:

Proof. We first show that T is linear. Consider $f, g \in P(\mathbb{R})$ and $c \in \mathbb{R}$. Then,

$$T(f(x) + g(x)) = \int_0^x f(t) dt + \int_0^x g(t) dt = \int_0^x f(t) + g(t) dt = T(f(x) + g(x))$$

$$T(cf(x)) = \int_0^x cf(t) dt = c \int_0^x f(t) dt = cT(f(x))$$

and hence T is linear. It is clear that T is not onto as any constant polynomial ($f(x) = n$ for $n \in \mathbb{R}$) is not in $R(T)$ since there is nothing that integrates to just a constant. We show that T is one-to-one by assuming that $T(f(x)) = T(g(x))$ and showing $f = g$:

$$\begin{aligned} T(f(x)) = T(g(x)) &\Rightarrow \int_0^x f(t) dt = \int_0^x g(t) dt \\ &\Rightarrow \int_0^x f(t) - g(t) dt = 0 \\ &\Rightarrow \frac{d}{dx} \left[\int_0^x f(t) - g(t) dt \right] = 0 \\ &\Rightarrow f(x) - g(x) = 0 \\ &\Rightarrow f(x) = g(x) \quad \forall x \in \mathbb{R} \end{aligned}$$

as was to be shown. □

2.2.2:

Solution:

(a) We have that

$$T(e_1) = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \quad \text{and} \quad T(e_2) = \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix}$$

so

$$[T]_{\beta}^{\gamma} = \begin{pmatrix} 2 & 1 \\ 3 & 4 \\ 1 & 0 \end{pmatrix} = A$$

and so the matrix representation fo T in the standard ordered bases for $\mathbb{R}^2$ and $\mathbb{R}^3$ is the same as the matrix presentation of T in the form $T = L_A$.

(b) We have that

$$T(e_1) = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \text{and} \quad T(e_2) = \begin{pmatrix} 3 \\ 0 \end{pmatrix} \quad \text{and} \quad T(e_3) = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

so

$$[T]_{\beta}^{\gamma} = \begin{pmatrix} 2 & 3 & -1 \\ 1 & 0 & 1 \end{pmatrix} = A$$

and so the matrix representation fo T in the standard ordered bases for $\mathbb{R}^2$ and $\mathbb{R}^3$ is the same as the matrix presentation of T in the form $T = L_A$.

(c) We have that

$$T(e_1) = 2 \quad \text{and} \quad T(e_2) = 1 \quad \text{and} \quad T(e_3) = -3$$

so

$$[T]_{\beta}^{\gamma} = \begin{pmatrix} 2 & 1 & -3 \end{pmatrix} = A$$

and so the matrix representation fo T in the standard ordered bases for $\mathbb{R}$ and $\mathbb{R}^3$ is the same as the matrix presentation of T in the form $T = L_A$.

2.2.4:

Solution: We have that

$$T(e_1) = 1 \quad \text{and} \quad T(e_2) = 1 + x^2 \quad \text{and} \quad T(e_3) = 0 \quad \text{and} \quad T(e_4) = x$$

so

$$[T]_{\beta}^{\gamma} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

2.2.5d:

Solution: We have that

$$T(e_1) = 1 \quad \text{and} \quad T(e_2) = 2 \quad \text{and} \quad T(e_3) = 4$$

$$\text{so } [T]_{\beta}^{\gamma} = \begin{pmatrix} 1 & 2 & 4 \end{pmatrix}$$